Would I Lie to You? Web App: Test Requirements

Approach

Every Must requirement in the BRD has at least one test below. Tests are split three ways:

Type	Tool	Covers
Unit	pytest	Game logic: scoring, round order, story/lie selection, sudden death, no-repeat rules, T/L validation
Integration	pytest + Flask test client, AI client mocked	Routes, session state, online/offline switching, fallback, history file
Manual	Test mode (FR23), checklist below	Screens, privacy step, Spacebar, timer, browsers, playtesting

- Unit and integration tests never need Ollama running. A fake AI client returns set responses (good lies, duplicates, offensive text, errors).
- The timer length is a config value so tests can run it at 5 seconds instead of 10 minutes.
- Random choices take a seed so selection tests are repeatable.
- The used-lies history file is pointed at a temporary file during tests.

Test cases

Grouped by type. Requirement IDs in brackets.

Unit tests (pytest)

ID	Test	Expected result
TC02 (FR3)	Play 6 rounds with scores not level	Game ends after round 6; teams alternate; each player speaks exactly once
TC06 (FR6)	Load the lie bank	At least 60 entries, no duplicates, none empty
TC09 (FR7)	Simulate 1,000 games with a fixed seed	Truth share 45 to 55%; lies per game varies (not always 3)
TC10 (FR7)	Run a full game plus sudden death	No true story is used twice
TC15 (FR12)	Inputs: t, T, l, L, x, empty, TL, 1	Only t/T/l/L accepted; others rejected with a message; nothing scores until Confirm
TC16 (FR13)	Correct guess, then wrong guess	Correct: guessing team +1. Wrong: speaker's team +1. Other team unchanged; no score below 0

ID	Test	Expected result
TC20 (FR22)	Scores 3-3 after 6 rounds; then a 1-1 pair; then a 2-0 pair	Sudden death starts; 1-1 pair repeats; 2-0 pair ends game with that team as winner
TC21 (FR19)	10 offline games in a row, every round forced to a lie	60 lies used, 0 duplicates
TC23 (FR19)	AI mock returns an old lie with one word changed	Flagged as near-duplicate and regenerated
TC24 (FR20)	Play games 11 to 20 offline	No bank entry repeats until every entry has been used once

Integration tests (pytest, AI mocked)

ID	Test	Expected result
TC01 (FR1, FR2)	Start setup with 2 players on a team, or a blank name	Blocked with a message; game only starts with 3 named players per team
TC03 (FR4)	Enter 1 story, an empty story, then 2 valid stories	First two blocked; 2 non-empty stories accepted; stories never shown on another player's screen
TC04 (FR5)	Local AI mode, mocked model returns valid lies	All lies ready before round 1; no AI call during rounds
TC05 (FR6)	AI mock raises an error / times out	Game still starts using the lie bank; message tells players offline lies are in use
TC07 (FR21)	Choose Lie bank on the setup screen	Game runs; AI client never called (assert mock not called)
TC08 (FR17)	AI mock returns an offensive lie	Lie rejected and regenerated; never reaches a speaker
TC19 (FR16)	End a game, click Play Again	End screen names the winner; Play Again clears scores, names and stories
TC22 (FR19)	AI mock returns a lie already in the history	Rejected and regenerated
TC25 (FR23)	Play a full game in test mode	TRUTH/LIE shown during each round; "TEST GAME" label shown; history file unchanged afterwards
TC30 (NFR5)	Time lie generation with the local model on the Mac, 5 runs	Every run under 15 s
TC31 (NFR6)	Finish a game, open the history file	Contains lies only; no player names or true stories
TC35 (FR24)	Choose Local AI with a mocked local model	Lies come from the local model and are tagged as local
TC36 (FR24)	Choose Local AI with Ollama not running	Game starts using the lie bank; message names the local AI model

Manual tests

ID	Test	Expected result
TC11 (FR8)	Go through the privacy screen	Statement hidden by default; shows only on Reveal; hidden again before Spacebar; never visible on the guessing screen
TC12 (FR9)	Press Spacebar with statement showing, then hidden, then again mid-round	Nothing while showing; starts round when hidden; second press does not restart timer
TC13 (FR10)	Start a round with timer set to 5 s, then real 10:00	Starts at set time, counts down, moves to guessing at 0:00; real timer shows 10:00
TC14 (FR11)	Click Ready to Guess at 7:30	Timer stops; guess screen opens
TC17 (FR14)	Finish a round with a correct and a wrong guess	Reveal shows the statement, TRUTH or LIE, and the team that got the point
TC18 (FR15)	Play 3 rounds	Scoreboard and "Round x of 6" update after every round
TC26 (FR18)	Let a timer reach 0:00	Sound plays once
TC27 (NFR1, NFR2)	Fresh clone, `pip install -r requirements.txt`, `python app.py`	Game loads at localhost with no other steps; backend is Python
TC28 (NFR3)	Deploy to Render or Railway with only config changes	Game playable at the public URL
TC29 (NFR4)	Search the repo and Git history for secrets; check `.gitignore`	No secrets in code or history; `.env` ignored
TC32 (NFR7)	Full game in Chrome, Safari and Edge on a laptop	No layout or input problems
TC33 (NFR8)	Run `pytest`	All tests pass; logic, AI service and routes are separate modules
TC34 (Success criteria)	Playtest 3 full Local AI games with real players	Guessers correct on fewer than 70% of rounds
TC37 (FR24)	Play a full game with Local AI on a Mac running Ollama	Lies are generated on the laptop; game completes with no API key

Entry and exit criteria

Entry (testing can start)

- Code runs locally with `python app.py`
- Lie bank file has at least 60 entries
- Fake AI client and test config (short timer, temp history file, fixed seed) in place

Exit (ready to demo or deploy)

- Every test linked to a Must requirement passes
- `pytest` runs green with no skipped tests
- No known bug that stops a full game, shows the statement to the guessing team, or scores wrongly
- Should and Could tests (TC08, TC24, TC26) passed or logged as known issues